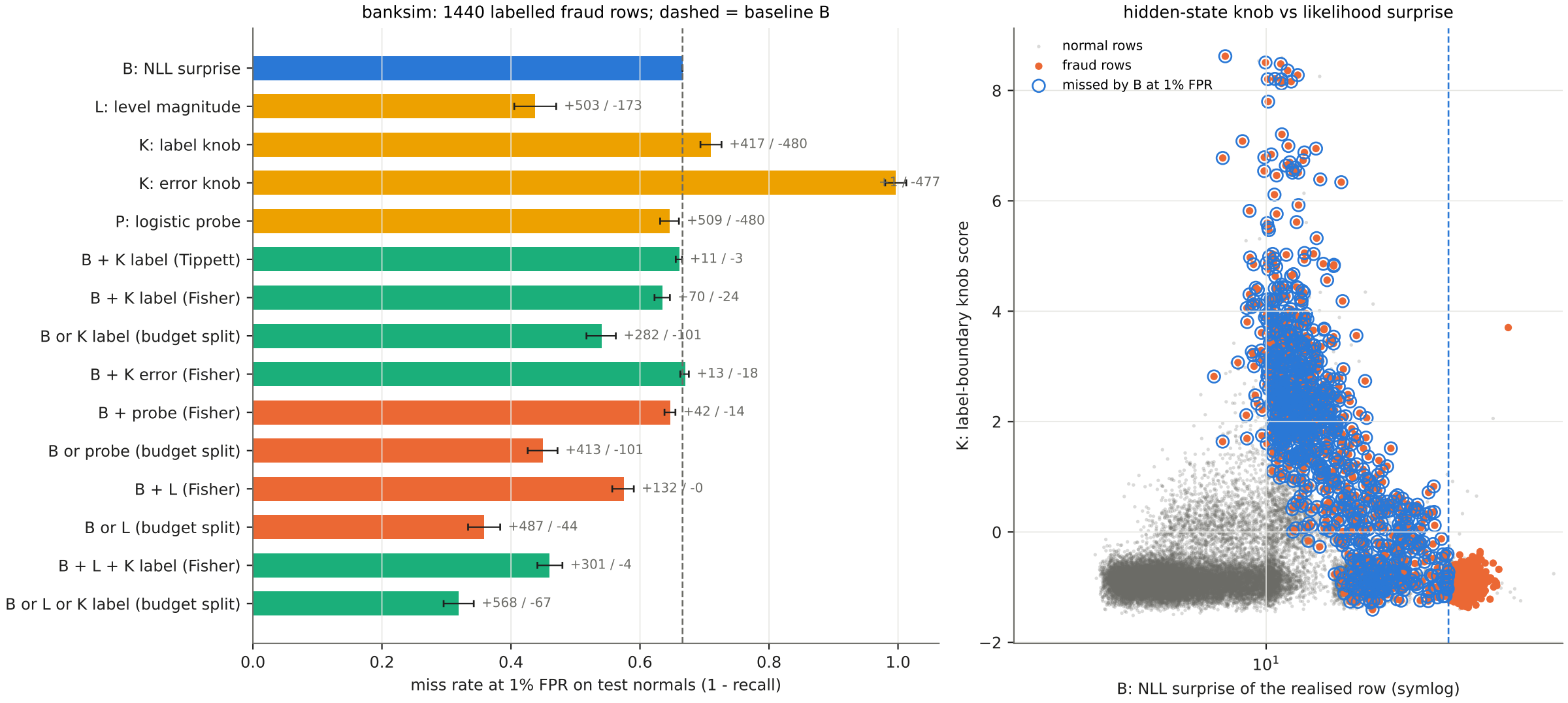

banksim: 1440 labelled fraud rows; dashed = baseline B
B: NLL surprise
L: level magnitude
+503 / -173
K: label knob
+417 / -480
K: error knob
-477
P: logistic probe
+509 / -480
B + K label (Tippett)
+11 / -3
B + K label (Fisher)
+70 / -24
B or K label (budget split)
+282 / -101
B + K error (Fisher)
+13 / -18
B + probe (Fisher)
+42 / -14
B or probe (budget split)
+413 / -101
B + L (Fisher)
+132 / -0
B or L (budget split)
+487 / -44
B + L + K label (Fisher)
+301 / -4
B or L or K label (budget split)
+568 / -67
0.0
0.2
0.4
0.6
0.8
1.0
miss rate at 1% FPR on test normals (1 - recall)
hidden-state knob vs likelihood surprise
normal rows
fraud rows
missed by B at 1% FPR
K: label-boundary knob score
8
6
4
2
0
−2
$10^1$
B: NLL surprise of the realised row (symlog)